

ELI5 How is ChatGPT bad for the environment?

Posted by u/CompetitivePickle831 • 0 points • 34 comments

I was listening to a podcast where they mentioned AI and ChatGPT. Didn't think anything of it until I went to the comments and it was dozens of people saying how bad ChatGPT is for the environment. I feel really guilty because I've used ChatGPT a lot but I would like to consider myself environmentally conscious. Can someone explain how it's bad for the environment??

Comments

u/HRudy94 • 2 points • 2025-07-08 15:08 UTC

The issue with ChatGPT (and others like LeChat, Gemini, Grok, Perplexity...) is that they require an excessive amount of power to create, maintain and even run.

As such, they use big datacenters plastered all over the world to run. Those are the actual issue and not the principle of LLMs/chat AIs themselves.

As opposed to traditional platforms, search engines etc that can at least utilize the CPU, AI models need to run on a graphics card for the raw processing speed. Those are more power-hungry and tend to heat a lot more.

So the AI-capable datacenters consume more energy, more water to cool down and emit more carbon emissions as the electronics are constantly under stress. Energy itself isn't really an issue if the country's energy source is decarbonated, the others though...

To put that into perspective, cloud gaming consumes either around the same or less than AI datacenters.

That said, you could get yourself a pretty powerful consumer-level GPU and run AI models locally on your machine for a much better carbon footprint overall. You're not gonna be running models as "smart" and powerful as ChatGPT's model, but you can get close enough. On top of that you get much better data privacy, since your requests never go to a greedy, spying company.

u/ScrivenersUnion • 2 points • 2025-07-08 15:00 UTC

There's a lot of misinformation going around right now on this subject, and a lot of information we simply don't have.

- * A few scientists have released energy use calculations, but these calculations were using a large and inefficient LLM model on an underpowered computer in their lab. Their results are likely much higher than reality and required quite a bit of assumption/extrapolation.
- * Anti-AI voices have repeated these studies and exaggerated them many, many times over. "Every ChatGPT query uses up 5 gallons of water" is a good example.
- * The actual companies themselves are not releasing information, likely because they don't want each other to know how efficient their models really are.
- * Many of the commercial AI models are not yet profitable, but that's also a poor indicator because they're flush with investment cash and haven't needed to prioritize this yet.
- * Open source LLM models available on HuggingFace and other sites can be remarkably efficient, and we've seen Ukrainian forces mount AI systems on drones - where things like battery use and weight are a premium.

Ultimately we don't know the truth and can only make extrapolations. Depending on how the calculating person views AI, their extrapolations can vary wildly.

But what we DO know for certain is that companies are paying a significant amount for server farms and investing in local power, cooling, and computation to meet those needs.

AI will likely get more efficient over time, but this will sit in equilibrium with how more sophisticated router-style multi agent models actually iterate multiple times over the same prompt.

u/badguy84 • 1 points • 2025-07-08 14:53 UTC

All services on the internet use energy, and LLMs (I will say this generically because this isn't a unique problem for ChatGPT) are no exceptions. There is a need for all the servers that hold the data with the model, and do the processing for your questions to be powered.

What makes LLMs in particular very power hungry is that basically what we did is: we trained a great chef with all the recipes in the entire world and access to the very best ingredients and techniques that exists. We ask it for a bowl of your favorite spaghetti and the chef digs in to all the world's recipes for spaghetti and all the known best ingredients and uses all the special techniques to make you a bowl of spaghetti. It's delicious but he used some weird herbs and different tomatoes from what you're used to so it tastes a little off.

Meanwhile your mom goes to the super market, gets some spaghetti and canned sauce and makes you an amazing bowl of spaghetti.

Now that chef has taken a ton of resources and uses a lot in order to make you that bowl of spaghetti. Your mom just went shopping and made it quickly. That's why people call out LLMs for being so power hungry. You are using a huge chunk of writing/art/photos/videos created by people all over the planet just to find the right answer for what you're asking. So that takes a lot of energy. If you instead would just use a calculator to calculate, or a dictionary to look up the meaning of the word it'd be far more efficient.

On top of all of this new data centers are being built as this AI is a bit of a boom and more computers are needed to run all of this stuff. So that puts additional emphasis on this as these data centers require power and almost get their own power plants. So that makes things stand out even more.

u/explainlikeimfive-ModTeam • 1 points • 2025-07-08 14:50 UTC

****Your submission has been removed for the following reason(s):****

Rule 7 states that users must search the sub before posting to avoid repeat posts within a year period. If your post was removed for a rule 7 violation, it indicates that the topic has been asked and answered on the sub within a short time span. Please search the sub before appealing the post.

If you would like this removal reviewed, please read the [detailed rules](https://www.reddit.com/r/explainlikeimfive/wiki/detailed_rules) first. ****If you believe this submission was removed erroneously****, please [use this form](<https://old.reddit.com/message/compose?to=%2Fr%2Fexplainlikeimfive&subject=Please%20review%20my%20thread?&message=Link:%20{url}%0A%0APlease%20answer%20the%20following%203%20questions:%20A%0A1.%20The%20concept%20I%20want%20explained:%20>

and we will review your submission.

u/Thatweasel • 1 points • 2025-07-08 14:50 UTC

Generative AI uses an outsized amount of electricity both in training and in running the model compared to other solutions i.e google searches. They're also being used more and more for tasks that are much more efficient and useful when manned by humans i.e customer support.

The data centers that handle running these models and the additional traffic they create use potable water for cooling which can create local water shortages.

Data centers are large and tend to be built in remote locations and have quite a large local environmental impact to build and maintain, and increased use from LLMs is driving the construction of more of them.

The hardware used by data centres is built with rare earth metals that require mining, which is also high environmental impact

u/wildfire393 • 1 points • 2025-07-08 14:46 UTC

Generative Neural Networks (the technology behind ChatGPT and most other "AI") works by running a huge number of calculations. This consumes electricity and produces heat, which must be dissipated, often using fresh water that then must be treated before it can be used again.

Consuming more electricity require more electricity being produced, which generally means continuing to use fossil fuels even as more green energy sources are added as the energy demand is increasing faster than the supply of green energy.

u/eemz53 • 1 points • 2025-07-08 14:44 UTC

It uses a ton of servers, which generate heat. The easiest way to offset that heat is to water. The heat evaporates the water, which usually is the same water that people in the area drink. Rain will replenish it eventually, but there is still a big reduction in available water to drink

u/eemz53 • 0 points • 2025-07-08 14:44 UTC

It uses too much energy to rely on solar, so they have to supplement with fossil fuels

u/soundman32 • 1 points • 2025-07-08 14:44 UTC

Each request uses around 40W/h* of electricity to run (even though it takes a couple of seconds to respond, it's 60 minutes of electricity burned). Multiply that by billions of requests each day, and you can see why it's bad for the electricity generators trying to keep up.

* Heard on a podcast. Equivalent to a large LED light running for an hour.

u/Ferociousfeind • -1 points • 2025-07-08 14:43 UTC

They use a bunch of energy, to train and to run, but... it's not an exceptional amount? Sure, it's more energy than we were using before, in exchange for the internet version of fast food (so basically no benefit whatsoever), but there are lots of more intense electricity-users with minimal benefit, like cryptocurrency mining rigs.

It's not much worse for the environment than many things we already do, but it IS pretty bad. It's bad because it uses a lot of electricity, and many of our techniques for generating electricity are bad for the environment